PROBLEM 27

Bead on a Rotating Sine-Shaped Wire with a Spring

Original statement · Problems.pdf, p. 15

Rotating Wire + Spring

A heavy bead P of mass m slides without friction along a thin wire with shape given by the following equation:

$$y_1(x_1) = \sin x_1.$$

Moreover, the bead is joined to a straight rail located at $y_1 = 2$ by a spring of stiffness k. The connection of the spring with the rail can move freely along the rail, so that the spring is always vertical. The whole setup rotates with respect to the y_1 axis with a constant angular velocity ω. A sketch of the setup and the definition of the right-handed, rotating $Ox_1y_1z_1$ reference frame is given below.

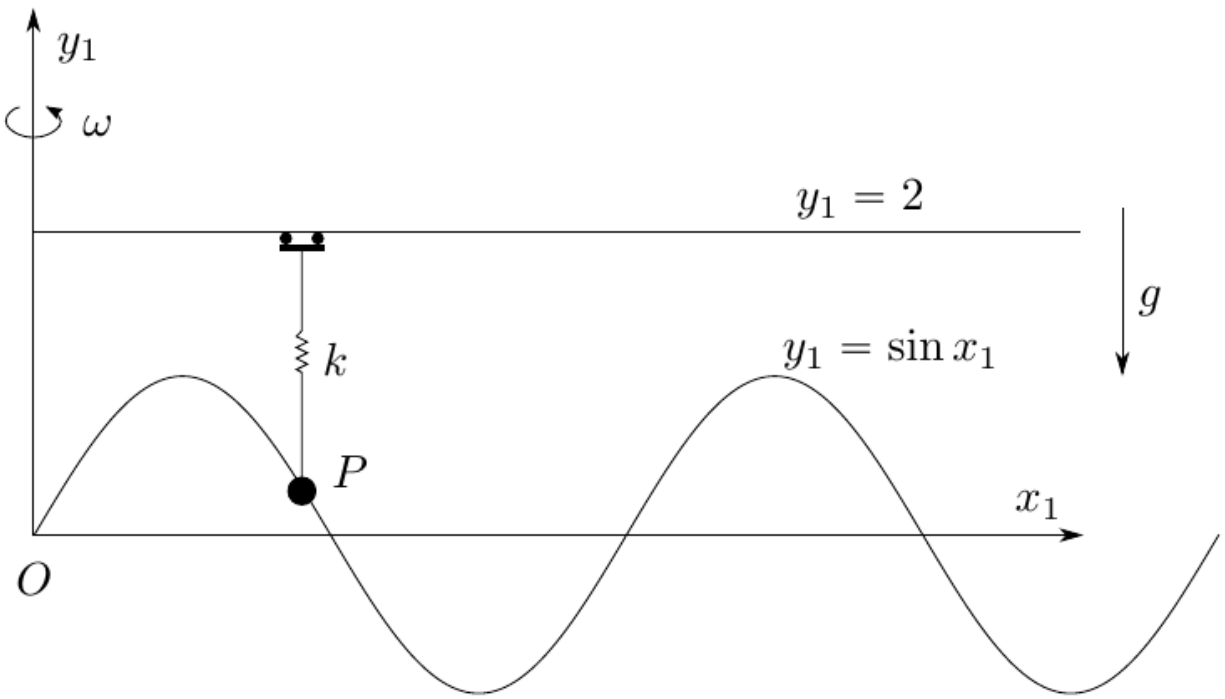


Figure: Sine-shaped wire y1 = sin x1 rotating about the vertical axis Oy1 with angular velocity ω; bead P joined by a vertical spring k to the rail y1 = 2 (Problems.pdf, p. 15).

(a) Write down the general expression for the components of the relative velocity and acceleration of P in the reference frame $Ox_1y_1z_1$ as functions of the axial position of the bead, x_1, and its time derivatives $\dot{x}_1$ and $\ddot{x}_1$.

(b) Identify all forces acting on the bead and then write down the three components of Newton's second law in the $Ox_1y_1z_1$ reference frame. From these equations, obtain an explicit expression of the components of the reaction force on the wire as a function of x_1, $\dot{x}_1$ and the parameters of the problem only, and a second order differential equation for x_1 that does not depend on the reaction force.

(c) Is it possible to use the equation of conservation of mechanical energy of the relative motion of P? If so, write it down in terms of x_1, $\dot{x}_1$ and the parameters of the problem. What is the relation of this equation with the differential equation for x_1 in the previous point?

(d) For $\omega = 0$ and $x_1 \in [0, 2\pi]$, discuss the existence of equilibrium positions as a function of the parameter $\lambda = mg/k$. Are these equilibrium positions stable?

(e) Consider now that $\omega \neq 0$, and $k = 0$. We drop the bead P at $x_1 = \pi$ with no initial velocity. Sketch the shape of the potential and discuss qualitatively the motion of P if the parameter $\beta = \omega^2/g$ is small or large. Write down the system of equations that would allow to solve for the critical value of β that divides between the two behaviors.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a heavy bead P (mass m) slides on the smooth plane wire $y_1 = \sin x_1$. A vertical spring of stiffness k joins it to the rail $y_1 = 2$; the upper end slides freely on the rail, so the spring stays vertical. Wire, rail and spring rotate about the vertical axis Oy_1 with constant angular velocity ω.
- **Hypotheses:** the non-rotating frame with origin O and vertical axis Oy_1 is inertial; gravity is $-g\,\mathbf{j}_1$; the wire is smooth (no reaction along the tangent). The statement does not give the natural length of the spring; it is taken as zero (the form of the official energy integral implies this), so the spring force is k times its length $2 - \sin x_1$. Lengths in the wire equation are measured in metres (the equation $y_1 = \sin x_1$ hides a unit length of 1 m).
- **Why the rotating frame:** in $Ox_1y_1z_1$ the wire, the rail and the spring are fixed, so the bead moves on a fixed curve with one coordinate x_1; the rotation enters only through the centrifugal and Coriolis forces (Notes Eq. 4.62).
- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$, with $\mathbf{j}_0$ vertical upwards.
- **Rotating frame:** $S_1 = Ox_1y_1z_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the wire: Ox_1 in the plane of the wire, Oy_1 the vertical axis of rotation ($\mathbf{j}_1 = \mathbf{j}_0$), Oz_1 normal to the plane of the wire; coincident with S_0 at $t = 0$.
- **Directions attached to the wire:** the tangent $\mathbf{t}$ (in the sense of increasing x_1), the in-plane normal $\mathbf{n}$ and the normal to the plane $\mathbf{k}_1$; deduced in the next section.
- **Degrees of freedom:** 3 coordinates minus 2 constraints ($z_1 = 0$ and $y_1 = \sin x_1$): CDOF $= 1$, generalized coordinate x_1.

Symbol	Meaning	SI unit
m	mass of the bead	kg
k	spring stiffness	N/m
ω	angular velocity of the setup (constant)	rad/s

Symbol	Meaning	SI unit
g	gravitational acceleration	$\mathrm{m/s^2}$
x_1	axial position of the bead	m
$\lambda = mg/k$	weight-to-stiffness ratio	m
$\beta = \omega^2/g$	centrifugal-to-gravity ratio	$\mathrm{m^{-1}}$

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

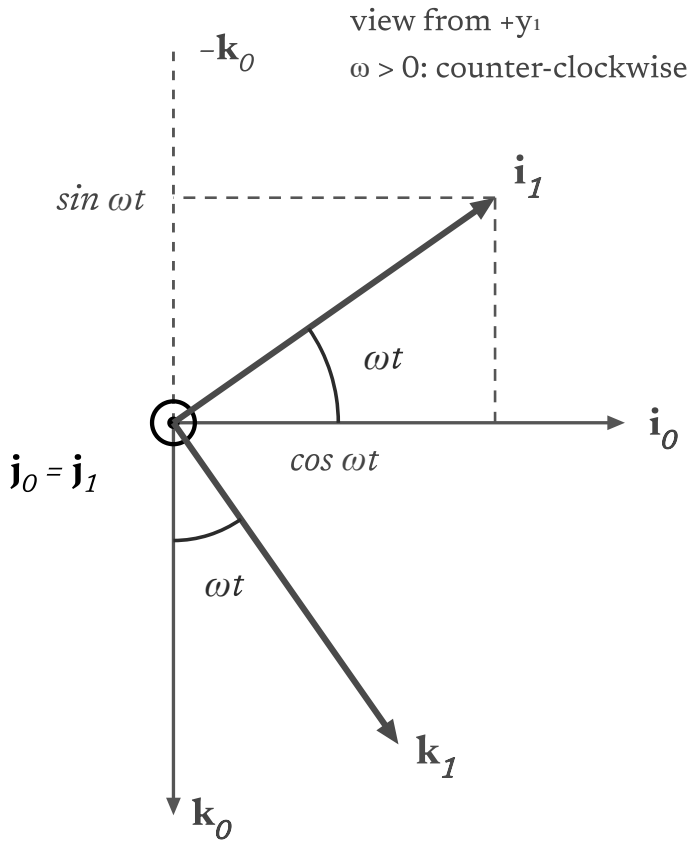


Figure 1. View from $+y_1$, with $\mathbf{j}_0 = \mathbf{j}_1$ out of the paper ($\odot$): $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$ points down the page. The plane of the wire turns counter-clockwise by ωt: the components of $\mathbf{i}_1$ are $\cos\omega t$ along $\mathbf{i}_0$ and $\sin\omega t$ along $-\mathbf{k}_0$ (up the page), hence $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 - \sin\omega t\,\mathbf{k}_0$.

The setup turns about $\mathbf{j}_0 = \mathbf{j}_1$, so the rotation is drawn looking down the axis, from $+y_1$: $\mathbf{j}_0$ points out of the paper, towards the reader ($\odot$). With $\mathbf{i}_0$ to the right, the right-hand rule $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$ (fingers from $\mathbf{i}_0$, to the right, towards $\mathbf{j}_0$, towards the reader) puts $\mathbf{k}_0$ down the page;

$-\mathbf{k}_0$ points up the page. With $\omega > 0$ the wire turns counter-clockwise in this view (thumb along $\mathbf{j}_0$, towards the reader), and after a time t the axis $\mathbf{i}_1$ has turned ωt from $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the dashed right triangle: the leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos\omega t$; the opposite leg points up the page, the direction $-\mathbf{k}_0$, and measures $\sin\omega t$. Hence $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,(-\mathbf{k}_0)$: **the minus sign comes from $\mathbf{k}_0$ pointing down the page in this view.**
- $\mathbf{k}_1$ starts on $\mathbf{k}_0$ (down the page) and turns the same angle counter-clockwise, so its tip moves towards $+\mathbf{i}_0$: adjacent leg $\cos\omega t$ along $\mathbf{k}_0$, opposite leg $+\sin\omega t$ along $\mathbf{i}_0$.
- $\mathbf{j}_1 = \mathbf{j}_0$, the axis of the rotation.

$$\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 - \sin\omega t\,\mathbf{k}_0, \qquad \mathbf{j}_1 = \mathbf{j}_0, \qquad \mathbf{k}_1 = \sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & 0 & \sin\omega t \\ 0 & 1 & 0 \\ -\sin\omega t & 0 & \cos\omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\omega t\cos\omega t - \sin\omega t\,(-\sin\omega t) = \cos^2\omega t + \sin^2\omega t = 1$$

Multiplying out, $[{}_0R_1][{}_0R_1]^T$ has diagonal entries $\cos^2\omega t + \sin^2\omega t = 1$ (and 1) and off-diagonal entries $\pm(\cos\omega t\sin\omega t - \sin\omega t\cos\omega t) = 0$, so it is the identity.

Motion of S_1. $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \omega\,\mathbf{j}_1$ (constant), $\boldsymbol{\alpha}_{10} = \mathbf{0}$. Check with Poisson's formula: by the chain rule, $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\omega \sin\omega t\,\mathbf{i}_0 - \omega\cos\omega t\,\mathbf{k}_0 = -\omega\,\mathbf{k}_1$, and $\omega\,\mathbf{j}_1 \times \mathbf{i}_1 = -\omega\,\mathbf{k}_1$.

Position vector, tangent and normal in the plane of the wire

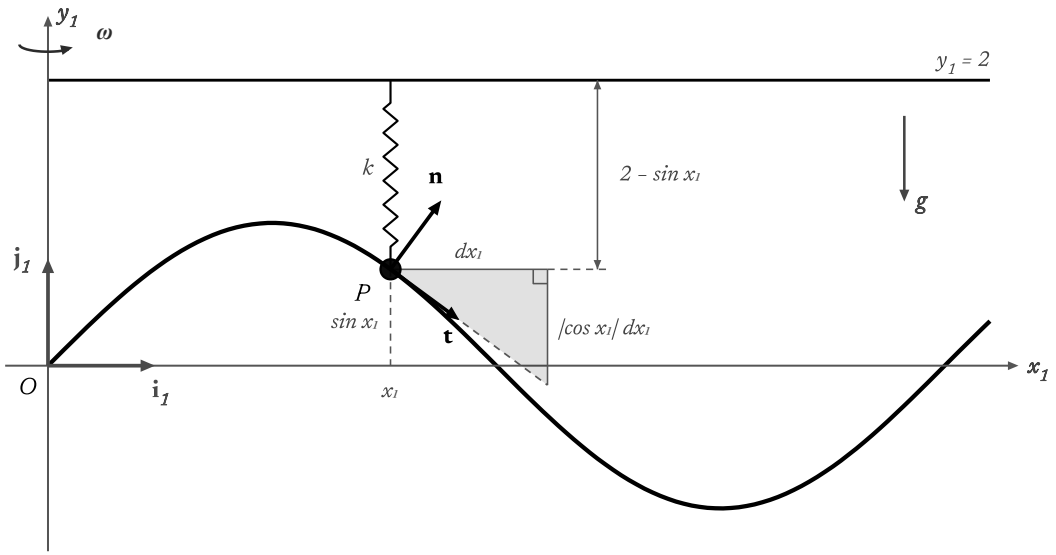


Figure 2. Plane of the wire, fixed in $Ox_1y_1z_1$. The dashed lines are the components x_1 and $\sin x_1$ of $\mathbf{r}_1^P$; the spring is vertical, of length $2 - \sin x_1$. In the shaded slope triangle the horizontal leg is dx_1 and the vertical leg is $dy_1 = \cos x_1\,dx_1$ (negative here: the wire goes down), so the tangent direction is $\mathbf{t} = \mathbf{i}_1 + \cos x_1\,\mathbf{j}_1$ and the in-plane normal is $\mathbf{n} = -\cos x_1\,\mathbf{i}_1 + \mathbf{j}_1$ (orange, drawn as unit vectors).

The plane of the wire is fixed in S_1 and is drawn with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up ($\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ out of the paper). The position of the bead is the sum of its two dashed components, the abscissa x_1 and the height given by the wire:

$$\mathbf{r}_1^P = x_1\,\mathbf{i}_1 + \sin x_1\,\mathbf{j}_1$$

The spring goes vertically from P to the rail $y_1 = 2$, so its length is the difference of heights, $2 - \sin x_1 > 0$ (always positive because $\sin x_1 \le 1$).

Slope triangle. By the chain rule, $dy_1 = \frac{d}{dx_1}(\sin x_1)\,dx_1 = \cos x_1\,dx_1$. In the shaded triangle the horizontal leg is dx_1 (along $\mathbf{i}_1$) and the vertical leg is $\cos x_1\,dx_1$ (along $\mathbf{j}_1$; drawn downwards because $\cos x_1 < 0$ at the position shown). The hypotenuse points along the wire, so a (not normalised) tangent vector is the vector sum of the two legs divided by dx_1, and its length follows from Pythagoras' theorem:

$$\mathbf{t} = \mathbf{i}_1 + \cos x_1\,\mathbf{j}_1, \qquad |\mathbf{t}|^2 = 1 + \cos^2 x_1$$

In-plane normal. Turning $\mathbf{t}$ by $+90^\circ$ about $\mathbf{k}_1$ (counter-clockwise in the drawing, since $\mathbf{k}_1$ points towards the reader) takes the components $(1, \cos x_1)$ to $(-\cos x_1, 1)$:

$$\mathbf{n} = \mathbf{k}_1 \times \mathbf{t} = -\cos x_1\,\mathbf{i}_1 + \mathbf{j}_1, \qquad \mathbf{t}\cdot\mathbf{n} = -\cos x_1 + \cos x_1 = 0$$

(with $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$). $\mathbf{t}$, $\mathbf{n}$ and $\mathbf{k}_1$ are mutually orthogonal; $\mathbf{n}$ has the same length as $\mathbf{t}$.

Kinematics

(a) Relative velocity and acceleration

Pedagogical Justification: $\mathcal{B}_1$ is constant in S_1, so the relative velocity and acceleration are the time derivatives of the components of $\mathbf{r}_1^P$ (Notes Eq. 3.7).

Chain rule: $\frac{d}{dt}\sin x_1(t) = \cos x_1\,\dot{x}_1$. Product and chain rule: $\frac{d}{dt}\left(\cos x_1\,\dot{x}_1\right) = -\sin x_1\,\dot{x}_1\cdot\dot{x}_1 + \cos x_1\,\ddot{x}_1 = -\sin x_1\,\dot{x}_1^2 + \cos x_1\,\ddot{x}_1$. Hence

$$\mathbf{v}_1^P = \dot{x}_1\,\mathbf{i}_1 + \cos x_1\,\dot{x}_1\,\mathbf{j}_1 + 0\,\mathbf{k}_1 = \dot{x}_1\,\mathbf{t}$$

$$\mathbf{a}_1^P = \ddot{x}_1\,\mathbf{i}_1 + \left(\cos x_1\,\ddot{x}_1 - \sin x_1\,\dot{x}_1^2\right)\mathbf{j}_1 + 0\,\mathbf{k}_1$$

Forces and Equations of Motion

(b) Forces on the bead, Newton's law, reaction and equation of motion

Pedagogical Justification: in the non-inertial frame S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63). Of the four inertia forces (Notes Eq. 4.62, Slide 28), only the centrifugal and the Coriolis forces survive, because $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$.

Real forces.

1. Weight: $\mathbf{W} = -mg\,\mathbf{j}_1$.
2. Spring: its length is $2 - \sin x_1 > 0$ and it pulls the bead up towards the rail: $\mathbf{F}_s = k\,(2 - \sin x_1)\,\mathbf{j}_1$.
3. Wire reaction: $\mathbf{N} = N_x\,\mathbf{i}_1 + N_y\,\mathbf{j}_1 + N_z\,\mathbf{k}_1$ with $\mathbf{N}\cdot\mathbf{t} = 0$, that is $N_x + N_y\cos x_1 = 0$ (smooth wire).

Inertia forces. With $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1, \mathbf{j}_1 \times \mathbf{j}_1 = \mathbf{0}, \mathbf{j}_1 \times \mathbf{k}_1 = \mathbf{i}_1$:

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \omega\,\mathbf{j}_1 \times (x_1\,\mathbf{i}_1 + \sin x_1\,\mathbf{j}_1) = -\omega x_1\,\mathbf{k}_1, \qquad \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = \omega\,\mathbf{j}_1 \times (-\omega x_1\,\mathbf{k}_1) = -\omega^2 x_1\,\mathbf{i}_1$$

$$\mathbf{F}_C = m\omega^2 x_1\,\mathbf{i}_1, \qquad \mathbf{F}_{COR} = -2m\,\omega\,\mathbf{j}_1 \times (\dot{x}_1\,\mathbf{i}_1 + \cos x_1\,\dot{x}_1\,\mathbf{j}_1) = 2m\omega\dot{x}_1\,\mathbf{k}_1$$

Three components of Newton's law in S_1:

$$\mathbf{i}_1: \quad m\ddot{x}_1 = N_x + m\omega^2 x_1$$

$$\mathbf{j}_1: \quad m\left(\cos x_1\,\ddot{x}_1 - \sin x_1\,\dot{x}_1^2\right) = N_y - mg + k\,(2 - \sin x_1)$$

$$\mathbf{k}_1: \quad 0 = N_z + 2m\omega\dot{x}_1$$

Out-of-plane reaction. From the $\mathbf{k}_1$ equation, $N_z = -2m\omega\dot{x}_1$: the wire cancels the Coriolis force.

In-plane reaction. Substitute $N_x = -N_y \cos x_1$ in the $\mathbf{i}_1$ equation, $m\ddot{x}_1 = -N_y \cos x_1 + m\omega^2 x_1$. Multiply it by $\cos x_1$ and substitute $m \cos x_1\, \ddot{x}_1$ in the $\mathbf{j}_1$ equation:

$$-N_y \cos^2 x_1 + m\omega^2 x_1 \cos x_1 - m \sin x_1\, \dot{x}_1^2 = N_y - mg + k\,(2 - \sin x_1)$$

Collect N_y:

$$N_y = \frac{m\left(\omega^2 x_1 \cos x_1 - \dot{x}_1^2 \sin x_1\right) + mg - k\,(2 - \sin x_1)}{1 + \cos^2 x_1}, \qquad N_x = -N_y \cos x_1$$

so the reaction is

$$\mathbf{N} = \frac{m\left(\omega^2 x_1 \cos x_1 - \dot{x}_1^2 \sin x_1\right) + mg - k\,(2 - \sin x_1)}{1 + \cos^2 x_1}\left(-\cos x_1\, \mathbf{i}_1 + \mathbf{j}_1\right) - 2m\omega\dot{x}_1\, \mathbf{k}_1$$

(the in-plane part is along $\mathbf{n}$, as it must be for a smooth wire).

Equation of motion. Project Newton's law on the tangent $\mathbf{t} = \mathbf{i}_1 + \cos x_1\, \mathbf{j}_1$: add the $\mathbf{i}_1$ equation and $\cos x_1$ times the $\mathbf{j}_1$ equation; $N_x + N_y \cos x_1 = 0$ removes the reaction:

$$m\ddot{x}_1 + m \cos x_1 \left(\cos x_1\, \ddot{x}_1 - \sin x_1\, \dot{x}_1^2\right) = m\omega^2 x_1 + \cos x_1\,[-mg + k\,(2 - \sin x_1)]$$

$$m\left(1 + \cos^2 x_1\right)\ddot{x}_1 - m \sin x_1 \cos x_1\, \dot{x}_1^2 = m\omega^2 x_1 - mg \cos x_1 + k \cos x_1\,(2 - \sin x_1)$$

Answers and Checks

(a) Relative velocity and acceleration

$\mathbf{v}_1^P = (\dot{x}_1,\ \cos x_1\,\dot{x}_1,\ 0)$ and $\mathbf{a}_1^P = \left(\ddot{x}_1,\ \cos x_1\,\ddot{x}_1 - \sin x_1\,\dot{x}_1^2,\ 0\right)$ in $\mathcal{B}_1$.

(b) Reaction and equation of motion

$N_x = -N_y \cos x_1$, $N_y = \dfrac{m\left(\omega^2 x_1 \cos x_1 - \dot{x}_1^2 \sin x_1\right) + mg - k\,(2 - \sin x_1)}{1 + \cos^2 x_1}$, $N_z = -2m\omega\dot{x}_1$; equation of motion $m\left(1 + \cos^2 x_1\right)\ddot{x}_1 - m \sin x_1 \cos x_1\,\dot{x}_1^2 = m\omega^2 x_1 - mg\cos x_1 + k\cos x_1\,(2 - \sin x_1)$.

(c) Conservation of the mechanical energy of the relative motion

> **Pedagogical Justification:** E_1^P is conserved if all the forces that do work in S_1 derive from a time-independent potential (Slide 29). Here $\boldsymbol{\omega}_{10}$ is constant and the wire is fixed in S_1.

Yes, it can be used. Check of each force: $\mathbf{N}\cdot\mathbf{v}_1^P = \dot{x}_1\,(N_x + N_y\cos x_1) = 0$; $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ (Slide 31); the weight has potential $mgy_1 = mg\sin x_1$; the spring has potential $\frac{1}{2}k\,(2 - y_1)^2 = \frac{1}{2}k\,(2 - \sin x_1)^2$ (its free end slides without friction, so only its length matters); the centrifugal force has potential $-\frac{1}{2}m\omega^2 r_\perp^2 = -\frac{1}{2}m\omega^2 x_1^2$ (Slide 30). With $T_1^P = \frac{1}{2}m|\mathbf{v}_1^P|^2 = \frac{1}{2}m\dot{x}_1^2\left(1 + \cos^2 x_1\right)$:

$$E_1^P = \tfrac{1}{2}m\dot{x}_1^2\left(1 + \cos^2 x_1\right) + mg\sin x_1 + \tfrac{1}{2}k\,(2 - \sin x_1)^2 - \tfrac{1}{2}m\omega^2 x_1^2 = \text{const}$$

Expanding $\frac{1}{2}k(2 - \sin x_1)^2 = 2k - 2k\sin x_1 + \frac{1}{2}k\sin^2 x_1$ and dropping the constant $2k$ gives the form of the official key, $\frac{1}{2}m\dot{x}_1^2(1 + \cos^2 x_1) + \sin x_1\,(mg - 2k) + \frac{1}{2}k\sin^2 x_1 - \frac{1}{2}m\omega^2 x_1^2 = C$.

Relation with the equation of motion. Differentiate E_1^P in time. Product and chain rules: $\frac{d}{dt}\left[\frac{1}{2}m\dot{x}_1^2(1 + \cos^2 x_1)\right] = m\dot{x}_1\ddot{x}_1(1 + \cos^2 x_1) - m\dot{x}_1^3 \sin x_1\cos x_1$, $\frac{d}{dt}(mg\sin x_1) = mg\cos x_1\,\dot{x}_1$, $\frac{d}{dt}\left[\frac{1}{2}k(2 - \sin x_1)^2\right] = -k(2 - \sin x_1)\cos x_1\,\dot{x}_1$, $\frac{d}{dt}\left(-\frac{1}{2}m\omega^2 x_1^2\right) = -m\omega^2 x_1\dot{x}_1$. Therefore

$$\frac{dE_1^P}{dt} = \dot{x}_1\left[m\left(1 + \cos^2 x_1\right)\ddot{x}_1 - m\sin x_1\cos x_1\,\dot{x}_1^2 - m\omega^2 x_1 + mg\cos x_1 - k\cos x_1\,(2 - \sin x_1)\right]$$

The bracket is exactly the equation of motion of (b), so $dE_1^P/dt = 0$ along every motion: the energy equation is a first integral of the equation of motion (obtained by multiplying it by $\dot{x}_1$ and integrating once), and the equation of motion is recovered by differentiating it (for $\dot{x}_1 \neq 0$).

(d) Equilibria for $\omega = 0$, $x_1 \in [0, 2\pi]$

> **Pedagogical Justification:** with $\omega = 0$ the equation of motion is $m(1+\cos^2 x_1)\ddot{x}_1 - m\sin x_1 \cos x_1 \dot{x}_1^2 = -dV/dx_1$, with $V(x_1) = mg\sin x_1 + \frac{1}{2}k(2-\sin x_1)^2$. Equilibria are the critical points of V; minima are stable, maxima unstable.

First derivative (chain rule):

$$V'(x_1) = mg\cos x_1 - k\,(2-\sin x_1)\cos x_1 = k\cos x_1\,(\lambda - 2 + \sin x_1)$$

Second derivative (product rule):

$$V''(x_1) = -k\sin x_1\,(\lambda - 2 + \sin x_1) + k\cos^2 x_1$$

Equilibria.

1. $\cos x_1 = 0$: $x_1 = \pi/2$ (crest of the wire) and $x_1 = 3\pi/2$ (trough). They exist for every λ.
2. $\sin x_1 = 2 - \lambda$: it has solutions only if $-1 \le 2-\lambda \le 1$, that is $1 \le \lambda \le 3$. For $1 < \lambda < 2$ they are $x_1 = \arcsin(2-\lambda)$ and $\pi - \arcsin(2-\lambda)$ (on both sides of the crest); for $\lambda = 2$, $x_1 = 0, \pi, 2\pi$; for $2 < \lambda < 3$, $x_1 = \pi + \arcsin(\lambda - 2)$ and $2\pi - \arcsin(\lambda - 2)$ (on both sides of the trough). At $\lambda = 1$ and $\lambda = 3$ they merge with $\pi/2$ and $3\pi/2$.

Stability.

- $x_1 = \pi/2$: $V''(\pi/2) = -k(\lambda - 1) = k(1-\lambda)$: stable for $\lambda < 1$ (a stiff spring holds the bead on the crest), unstable for $\lambda > 1$.
- $x_1 = 3\pi/2$: $V''(3\pi/2) = +k(\lambda - 3)$: stable for $\lambda > 3$ (a heavy bead sits in the trough), unstable for $\lambda < 3$.
- $\sin x_1 = 2 - \lambda$ ($1 < \lambda < 3$): the bracket vanishes, so $V'' = k\cos^2 x_1 > 0$: stable.
- Boundary cases: for $\lambda = 1$ near $\pi/2$ ($u = x_1 - \pi/2$) and for $\lambda = 3$ near $3\pi/2$ ($u = x_1 - 3\pi/2$), the Taylor expansion gives $V' \approx \frac{1}{2}ku^3$, so V has a quartic minimum: stable, but with zero linear frequency.

Discrepancy with the official solution

The official key (Problems.pdf, PDF page 74) states the results in terms of a parameter α: "$\pi/2$ stable if $\alpha > 1$; $3\pi/2$ stable if $\alpha < 1/3$; $\sin x_1 = 2 - 1/\alpha$, which exists and is stable if $1/3 < \alpha < 1$". These coincide with the results above only if $\alpha = 1/\lambda = k/(mg)$, the reciprocal of the parameter $\lambda = mg/k$ defined in the statement. Check: at $\pi/2$, $V''(\pi/2) = k(1-\lambda) > 0 \iff \lambda < 1 \iff 1/\lambda > 1$. The key also writes the in-plane reaction direction as $(-\cos x\,\mathbf{i}_1 + \mathbf{j}_1)$, a typo for $(-\cos x_1\,\mathbf{i}_1 + \mathbf{j}_1)$; its reaction formulas and energy integral otherwise agree with (b) and (c).

(e) $\omega \neq 0$, $k = 0$: bead dropped at $x_1 = \pi$

Pedagogical Justification: the energy integral of (c) with $k = 0$ is $\frac{1}{2}m\dot{x}_1^2(1+\cos^2 x_1) + V(x_1) = V(\pi)$, so the bead can only be where $V(x_1) \leq V(\pi)$; the motion is read from the shape of V (turning points where $V = V(\pi)$).

Dimensionless potential. With $k = 0$, $V = mg\sin x_1 - \frac{1}{2}m\omega^2 x_1^2$. Define $\tilde{V} = 2V/(mg)$:

$$\tilde{V}(x_1) = 2\sin x_1 - \beta x_1^2, \qquad \beta = \frac{\omega^2}{g}, \qquad \tilde{V}(\pi) = -\beta\pi^2$$

and the energy equation becomes $\dfrac{\dot{x}_1^2}{g}\left(1+\cos^2 x_1\right) = \tilde{V}(\pi) - \tilde{V}(x_1)$.

Shape of the potential (sketch). $\tilde{V}$ is a sine wave of amplitude 2 riding on the downward parabola $-\beta x_1^2$: wells near the troughs $x_1 \approx 3\pi/2, 7\pi/2, \ldots$, humps near the crests $x_1 \approx 5\pi/2, 9\pi/2, \ldots$, and $\tilde{V} \to -\infty$ as $|x_1| \to \infty$. The humps flatten and disappear as β grows, because $\tilde{V}' = 2\cos x_1 - 2\beta x_1$ can vanish only where $|\beta x_1| \leq 1$. For example, $\beta = 0.02\ \mathrm{m}^{-1}$: $\tilde{V}(\pi) = -0.197$ m, $\tilde{V}(3\pi/2) = -2.44$ m, $\tilde{V}(5\pi/2) = +0.77$ m; $\beta = 0.1\ \mathrm{m}^{-1}$: $\tilde{V}(\pi) = -0.99$ m, $\tilde{V}(5\pi/2) = -4.17$ m.

Start of the motion. $\tilde{V}'(\pi) = 2\cos\pi - 2\beta\pi = -2(1+\beta\pi) < 0$, so the generalized force $-V'$ is positive: the bead starts moving towards larger x_1 (down the wire into the trough near $3\pi/2$), and $x_1 = \pi$ is a turning point.

Small β (slow rotation). The hump near $5\pi/2$ is higher than the starting level, $\tilde{V} > -\beta\pi^2$ there, so the bead stops at the turning point x_t with $\tilde{V}(x_t) = \tilde{V}(\pi)$ and returns: it oscillates periodically between $x_1 = \pi$ and x_t (in the example $\beta = 0.02\ \mathrm{m}^{-1}$, $0.77 > -0.197$).

Large β (fast rotation). The centrifugal potential dominates, no hump to the right of π reaches the starting level, $\tilde{V}(x_1) < \tilde{V}(\pi)$ for all $x_1 > \pi$, and the bead moves outwards without bound ($x_1 \to \infty$) (in the example $\beta = 0.1$ m^{-1}, $-4.17 < -0.99$).

Critical value. The two behaviours are separated when the first hump to the right of π has its maximum exactly at the starting level: at that x_c the potential is stationary and equal to $\tilde{V}(\pi)$. The system of equations for (x_c, β_c) is

$$\begin{cases} \tilde{V}(x_c) = 2\sin x_c - \beta_c x_c^2 = -\beta_c \pi^2 \\ \tilde{V}'(x_c) = 2\cos x_c - 2\beta_c x_c = 0 \end{cases}$$

with the condition $\tilde{V}''(x_c) = -2\sin x_c - 2\beta_c < 0$ (a maximum). Solving it numerically (extension: the statement asks only for the system) gives $x_c \approx 2.401\pi \approx 7.544$ m and $\beta_c \approx 0.0405$ m^{-1}. Hence the bead oscillates for $\beta < \beta_c$ and escapes for $\beta > \beta_c$.

> **Discrepancy with the official solution**
> The official key writes the same system of equations for x_c and β_c, but states "$\beta > \beta_c$, the bead oscillates; $\beta < \beta_c$, the bead moves towards infinity". This is reversed. Proof: the centrifugal term $-\beta x_1^2$ is the only one that pushes the bead outwards, so a larger β lowers the humps; in the extreme $\beta \to \infty$, $\tilde{V}' = 2\cos x_1 - 2\beta x_1 < 0$ for all $x_1 \geq \pi$ (no hump at all), while for $\beta \to 0$ the hump at $5\pi/2$ has $\tilde{V} \to 2 > \tilde{V}(\pi) \to 0$. Numerical integration of the equation of motion (sympy) gives a bounded oscillation $x_1 \in [\pi, 2.31\pi]$ for $\beta = 0.95\beta_c$ and escape for $\beta = 1.05\beta_c$.

Checks

- **Role of each force.** The spring pulls the bead towards the rail, which on this wire favours the crest $\pi/2$; the weight favours the trough $3\pi/2$; the ratio $\lambda = mg/k$ decides, with the intermediate equilibria $\sin x_1 = 2 - \lambda$ where the vertical forces mg and $k(2 - \sin x_1)$ balance exactly. The centrifugal force pushes the bead away from the axis and, for fast rotation, throws it out; the Coriolis force only loads the wire (N_z) and never changes the energy.
- **Limits.** $\omega \to 0$: $N_z \to 0$ and the equation of motion is that of a bead on a fixed wire. $k \to 0$ in (d): $\lambda \to \infty$, only $3\pi/2$ is stable (bead in the trough). $k \to \infty$: $\lambda \to 0$, only $\pi/2$ is stable.
- **Orthogonality:** $\mathbf{t} \cdot \mathbf{n} = 0$ and $\mathbf{t} \cdot \mathbf{k}_1 = \mathbf{n} \cdot \mathbf{k}_1 = 0$; $[{}_0R_1]$ is orthogonal with $\det = 1$.

- **Dimensions** (with the implicit unit length of the wire): $[m\omega^2 x_1] = [mg] = [k(2 - \sin x_1)] = \mathrm{N}$; $[m\omega\dot{x}_1] = \mathrm{kg}\cdot\mathrm{s}^{-1}\cdot\mathrm{m}\cdot\mathrm{s}^{-1} = \mathrm{N}$; every term of E_1^P is in J ($[k(2 - \sin x_1)^2] = \mathrm{N/m}\cdot\mathrm{m}^2$); $[\lambda] = \mathrm{m}$ is compared with lengths $(2 - \sin x_1)$; $[\beta x_1^2] = \mathrm{m}^{-1}\cdot\mathrm{m}^2 = \mathrm{m} = [2\sin x_1]$.
- **Symbolic and numerical check (sympy):** $[{}_0R_1]$; $\mathbf{v}_1^P$, $\mathbf{a}_1^P$; Notes Eq. 3.27 against differentiation in $\mathcal{B}_0$; $\mathbf{F}_C$, $\mathbf{F}_{COR}$; N_x, N_y, N_z and the equation of motion solved from the three components (and again from $m\mathbf{a}_0^P = \mathbf{F}$ in S_0); $dE_1^P/dt = \dot{x}_1\times$(equation of motion); V', V'' and the boundary cases; the critical point (x_c, β_c) and the two behaviours by numerical integration: all verified.
- **Check against the official key:** the key of `Problems.pdf` (PDF page 74) gives the reaction, the energy integral and the equilibria, which agree with (b), (c) and (d) up to the two points documented in the discrepancy notes above (the parameter $\alpha = 1/\lambda$ of the stability statements and the reversed inequality for β_c); both are proved there.

Source: Problems.pdf (PDF page 15); key on PDF page 74